

PORTKEY Food Delivery System

Software Requirements Specification

Project: PORTKEY

Document: SRS

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PORTKEY Food Delivery System

E-commerce Platform for Online Food Ordering

This document specifies the software requirements for PORTKEY Food Delivery System based on IEEE 830-1998 standard. The system enables users to order food online from local restaurants, make payments, track deliveries, and interact with an intelligent chatbot for customer support.

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Revision History

Name	Date	Reason For Changes	Version
Team PORTKEY	September 2, 2025	Initial SRS Document based on IEEE 830-1998	1.0

1. Introduction

1.1 Purpose

The main purpose of this document is to specify the software requirements for the PORTKEY Food Delivery System based on IEEE 830-1998 standard. This document provides a comprehensive description of the system's functional and non-functional requirements, interfaces, and design constraints.

1.2 Scope

This document specifies requirements for PORTKEY Food Delivery System, an e-commerce platform that enables users to order food online from local restaurants.

The application allows users to:

- Browse local restaurants and their menus
- Add items to shopping cart and place orders
- Make payments using COD or online methods
- Track delivery status in real-time
- Communicate with delivery partners
- Interact with ML-powered chatbot for support
- View order history and manage profiles
- Rate and review restaurants
- Receive notifications for order updates

The system includes an admin dashboard for restaurant owners and system administrators to manage orders, menus, and user accounts. The application provides real-time order tracking and automated customer support through machine learning chatbot.

Note: The application functionality corresponds to a comprehensive food delivery platform with integrated ML support and multi-user management capabilities.

1.3 Product perspective

1.3.1 System interfaces

The application runs as a web-based platform compatible with modern browsers (Chrome, Firefox, Safari, Edge) on Windows, Linux, Mac, and mobile devices (Android/iOS browsers).

1.3.2 User interfaces

The application GUI provides responsive web interface with menus, navigation bars, forms, cards, and interactive elements allowing for easy control by keyboard, mouse, and touch gestures on mobile devices.

1.3.3 Hardware interfaces

Development Hardware Requirements:

- Intel i3/i5 processor minimum
- 8GB RAM minimum
- 500GB hard disk space
- Stable broadband internet connection

1.3.4 Software interfaces

The application integrates with external services via APIs:

- **Payment Gateways:** Integration for secure online transactions
- **SMS/Email Services:** For order notifications and confirmations
- **Cloud Storage:** For data synchronization and backup
- **Mapping Services:** For delivery tracking and location services
- **Restaurant POS Systems:** For menu updates and order management

Data formats supported:

- **REST APIs:** JSON-based web services
- **Database:** MySQL/MongoDB for data storage
- **File Formats:** Support for image uploads (JPEG, PNG)

1.3.5 Communications interfaces

- **HTTPS Protocol:** Secure communication for all data transmission
- **WebSocket:** Real-time order tracking updates
- **REST APIs:** Client-server communication
- **SMS Gateway:** Order status notifications
- **Email SMTP:** Registration and order confirmations

1.3.6 Memory constraints

The system shall efficiently manage memory usage to support concurrent users and real-time operations without performance degradation.

1.3.7 Operations

The system operates 24/7 with scheduled maintenance windows during low-traffic hours. Automated backup procedures and monitoring systems ensure continuous operation.

1.3.8 Site adaptation requirements

The system shall adapt to different geographic locations with support for local currencies, languages, and regional food preferences.

1.3.9 Interfaces with services

Integration with third-party services including payment processors, notification services, and mapping APIs for comprehensive functionality.

1.4 Product functions

Product functions include user management, restaurant browsing, order processing, payment handling, delivery tracking, chatbot support, and administrative management as detailed in Section 2.2 Functions.

1.5 User characteristics

Primary Users:

- **Customers:** General public with basic smartphone/computer literacy, familiar with e-commerce platforms
- **Restaurant Owners:** Business users comfortable with web applications for menu and order management
- **Delivery Partners:** Mobile-savvy individuals using the system for delivery coordination
- **System Administrators:** Technical users managing overall system operations

User Expertise Levels:

- **Novice:** Basic web navigation skills
- **Intermediate:** Comfortable with online shopping and payments
- **Expert:** Restaurant owners and admins with business application experience

1.6 Limitations

- Internet connection required for all operations
- Compatible with modern browsers only (IE not supported)
- Real-time features require stable internet connectivity
- Mobile responsiveness limited to standard screen sizes
- Payment processing dependent on third-party gateway availability

1.7 Assumptions and dependencies

Assumptions:

- Users have reliable internet connectivity
- Restaurants maintain updated menu information
- Delivery partners are available during operating hours
- Payment gateways maintain 99.9% uptime
- Users possess valid email addresses and phone numbers

Dependencies:

- Third-party payment gateway services
- Cloud hosting infrastructure reliability
- Restaurant cooperation for menu updates
- SMS/Email service provider availability
- Machine learning libraries for chatbot functionality

1.8 Definitions

Cart: Virtual shopping basket containing selected menu items before order placement.

Chatbot: Automated conversational agent using machine learning for customer support and query resolution.

Delivery Partner: Individual responsible for delivering orders from restaurant to customer location.

Order Status: Current state of an order (placed, preparing, dispatched, delivered, cancelled).

Real-time Tracking: Live updates on order preparation and delivery progress with location information.

Restaurant Listing: Catalog of available restaurants with menu information, ratings, and contact details.

1.9 Acronyms and abbreviations

API: Application Programming Interface

COD: Cash on Delivery

CRUD: Create, Read, Update, Delete

CSV: Comma Separated Values

GUI: Graphical User Interface

HTML: HyperText Markup Language

HTTPS: HyperText Transfer Protocol Secure

ID: Identifier

JSON: JavaScript Object Notation

LLM: Large Language Model

ML: Machine Learning

REST: Representational State Transfer

SQL: Structured Query Language

SRS: Software Requirements Specification

UI: User Interface

2. Requirements

2.1 External interfaces

External interfaces are described in Section 1.3 Product perspective including system, user, hardware, software, and communication interfaces.

2.2 Functions

2.2.1 User Management

2.2.1.1 User Registration

[PORTKEY-SRS-1] The application shall allow new users to register with email, password, phone number, and contact information.

[PORTKEY-SRS-2] The system shall validate email format and phone number during registration.

[PORTKEY-SRS-3] The system shall send email verification link upon successful registration.

2.2.1.2 User Authentication

[PORTKEY-SRS-4] The application shall allow users to login with email and password credentials.

[PORTKEY-SRS-5] The system shall authenticate users through secure login mechanism with encrypted password storage.

[PORTKEY-SRS-6] The application shall provide "Remember Me" option for user convenience.

2.2.1.3 Password Management

[PORTKEY-SRS-7] The application shall allow users to reset password via email verification.

[PORTKEY-SRS-8] The system shall enforce strong password requirements (minimum 8 characters, special characters).

2.2.1.4 Profile Management

[PORTKEY-SRS-9] The application shall allow users to view and update their profile information.

[PORTKEY-SRS-10] The system shall allow users to add multiple delivery addresses.

[PORTKEY-SRS-11] The application shall allow users to update contact information and preferences.

2.2.2 Restaurant Management

2.2.2.1 Restaurant Listing

[PORTKEY-SRS-12] The application shall display a list of available restaurants with basic information including name, cuisine type, ratings, and estimated delivery time.

[PORTKEY-SRS-13] The system shall show restaurant operating hours and availability status.

2.2.2.2 Menu Display

[PORTKEY-SRS-14] The application shall display restaurant menus with item names, descriptions, prices, and availability status.

[PORTKEY-SRS-15] The system shall organize menu items by categories (appetizers, main course, desserts, beverages).

[PORTKEY-SRS-16] The application shall display high-quality images for menu items where available.

2.2.2.3 Restaurant Search and Filter

[PORTKEY-SRS-17] The application shall allow users to search restaurants by name or cuisine type.

[PORTKEY-SRS-18] The system shall provide filtering options by cuisine type, ratings, delivery time, and price range.

[PORTKEY-SRS-19] The application shall allow sorting restaurants by popularity, ratings, delivery time, and cost.

2.2.3 Order Management

2.2.3.1 Shopping Cart

[PORTKEY-SRS-20] The application shall allow users to add menu items to shopping cart with quantity selection.

[PORTKEY-SRS-21] The system shall allow users to modify cart contents including adding, removing, and updating quantities.

[PORTKEY-SRS-22] The application shall display cart summary with itemized list, quantities, and total amount.

[PORTKEY-SRS-23] The system shall calculate total order amount including item cost, taxes, and delivery charges.

2.2.3.2 Order Placement

[PORTKEY-SRS-24] The application shall allow users to place orders with delivery address specification.

[PORTKEY-SRS-25] The system shall generate unique order IDs for tracking purposes.

[PORTKEY-SRS-26] The application shall allow users to add special instructions for orders.

[PORTKEY-SRS-27] The system shall provide order confirmation with estimated delivery time.

2.2.3.3 Order History

[PORTKEY-SRS-28] The application shall maintain complete order history for registered users.

[PORTKEY-SRS-29] The system shall allow users to reorder from previous orders.

[PORTKEY-SRS-30] The application shall display order details including items, prices, and delivery information.

2.2.4 Payment Processing

2.2.4.1 Payment Methods

[PORTKEY-SRS-31] The application shall support Cash on Delivery (COD) payment method.

[PORTKEY-SRS-32] The system shall integrate with online payment gateways for card payments.

[PORTKEY-SRS-33] The application shall support multiple payment options including credit cards, debit cards, and digital wallets.

2.2.4.2 Payment Processing

[PORTKEY-SRS-34] The system shall process payments securely with encryption and PCI DSS compliance.

[PORTKEY-SRS-35] The application shall provide payment confirmation and digital receipt generation.

[PORTKEY-SRS-36] The system shall handle payment failures gracefully with retry options and alternative payment methods.

2.2.5 Delivery Tracking

2.2.5.1 Order Status Updates

[PORTKEY-SRS-37] The application shall provide real-time order status updates including placed, confirmed, preparing, dispatched, out for delivery, and delivered.

[PORTKEY-SRS-38] The system shall send push notifications for order status changes.

[PORTKEY-SRS-39] The application shall display estimated delivery time and updates.

2.2.5.2 Delivery Partner Management

[PORTKEY-SRS-40] The system shall automatically assign delivery partners to confirmed orders.

[PORTKEY-SRS-41] The application shall provide contact information for assigned delivery partner.

[PORTKEY-SRS-42] The system shall allow communication between customers and delivery partners through in-app messaging or calling.

2.2.5.3 Live Tracking

[PORTKEY-SRS-43] The application shall provide real-time location tracking of delivery partner on map.

[PORTKEY-SRS-44] The system shall update delivery location every 30 seconds during active delivery.

2.2.6 Chatbot Functionality

2.2.6.1 ML-Powered Support

[PORTKEY-SRS-45] The application shall provide an ML-powered chatbot for customer queries and support.

[PORTKEY-SRS-46] The chatbot shall handle order inquiries, delivery status questions, and general FAQs.

[PORTKEY-SRS-47] The system shall provide 24×7 automated customer support through the chatbot.

2.2.6.2 Query Handling

[PORTKEY-SRS-48] The chatbot shall understand natural language queries and provide relevant responses.

[PORTKEY-SRS-49] The system shall escalate complex queries to human support when chatbot cannot resolve them.

[PORTKEY-SRS-50] The application shall maintain chat history for user reference and support continuity.

2.2.7 Rating and Review System

2.2.7.1 Restaurant Reviews

[PORTKEY-SRS-51] The application shall allow users to rate restaurants on a 5-star scale.

[PORTKEY-SRS-52] The system shall allow users to write detailed reviews for restaurants.

[PORTKEY-SRS-53] The application shall display average ratings and review summaries for restaurants.

2.2.7.2 Order Feedback

[PORTKEY-SRS-54] The system shall prompt users to rate their delivery experience after order completion.

[PORTKEY-SRS-55] The application shall collect feedback on food quality, delivery time, and overall experience.

2.2.8 Admin Dashboard

2.2.8.1 Order Management

[PORTKEY-SRS-56] The admin dashboard shall provide comprehensive view of all orders with filtering and search capabilities.

[PORTKEY-SRS-57] The system shall allow administrators to update order status and manage order exceptions.

[PORTKEY-SRS-58] The application shall provide order analytics including daily, weekly, and monthly reports.

2.2.8.2 User Management

[PORTKEY-SRS-59] The admin dashboard shall allow viewing and managing user accounts.

[PORTKEY-SRS-60] The system shall provide user activity reports and analytics.

[PORTKEY-SRS-61] The application shall allow administrators to handle user complaints and support requests.

2.2.8.3 Restaurant Management

[PORTKEY-SRS-62] The admin dashboard shall allow management of restaurant information, menus, and availability.

[PORTKEY-SRS-63] The system shall provide restaurant performance analytics including orders, ratings, and revenue.

2.2.9 Notification System

2.2.9.1 Push Notifications

[PORTKEY-SRS-64] The application shall send push notifications for order confirmations, status updates, and delivery alerts.

[PORTKEY-SRS-65] The system shall allow users to customize notification preferences.

2.2.9.2 Email and SMS

[PORTKEY-SRS-66] The system shall send email confirmations for account registration and order placement.

[PORTKEY-SRS-67] The application shall send SMS notifications for critical order updates and delivery confirmations.

2.3 Usability requirements

[PORTKEY-SRS-68] The application shall provide intuitive user interface design with clear navigation and menu structure.

[PORTKEY-SRS-69] The system shall be accessible to users with disabilities following WCAG 2.1 guidelines.

[PORTKEY-SRS-70] The application shall provide help documentation and contextual tooltips for user guidance.

[PORTKEY-SRS-71] The system shall support multi-language capability for broader user accessibility.

2.4 Performance requirements

2.4.1 Response Time

[PORTKEY-SRS-72] The application shall load the main page within 3 seconds on standard broadband connections.

[PORTKEY-SRS-73] The system shall process order placement within 5 seconds of user confirmation.

[PORTKEY-SRS-74] The chatbot shall respond to user queries within 2 seconds for standard inquiries.

2.4.2 Concurrent Users

[PORTKEY-SRS-75] The system shall support concurrent access for up to 1000 simultaneous users without performance degradation.

[PORTKEY-SRS-76] The application shall handle peak traffic during meal times (12-2 PM, 7-9 PM) efficiently.

2.4.3 System Availability

[PORTKEY-SRS-77] The system shall maintain 99.5% uptime during operating hours (6 AM - 12 AM).

[PORTKEY-SRS-78] The application shall provide graceful degradation during high load conditions.

2.4.4 Database Performance

[PORTKEY-SRS-79] Database queries shall execute within 1 second for standard operations (login, menu loading, order placement).

[PORTKEY-SRS-80] The system shall handle up to 10,000 concurrent database connections efficiently.

2.5 Logical database requirements

[PORTKEY-SRS-81] The application shall store user data including UserID, Name, Email, Password (encrypted), Phone, Address, and Role.

[PORTKEY-SRS-82] The system shall maintain restaurant data including RestaurantID, Name, Address, Contact, Operating Hours, and Cuisine Type.

[PORTKEY-SRS-83] The database shall store menu information with ItemID, RestaurantID, Name, Description, Price, Category, and Availability.

[PORTKEY-SRS-84] The system shall maintain order data including OrderID, UserID, RestaurantID, Items, TotalAmount, Status, and Timestamp.

[PORTKEY-SRS-85] The application shall store delivery partner information including PartnerID, Name, Phone, Status, and Current Orders.

[PORTKEY-SRS-86] The database shall enforce referential integrity with proper foreign key constraints and data validation.

[PORTKEY-SRS-87] The system shall encrypt sensitive information including passwords and payment data.

2.6 Design constraints

[PORTKEY-SRS-88] The system shall use responsive web design principles for mobile compatibility across devices.

[PORTKEY-SRS-89] The frontend shall be developed using HTML5, CSS3, and JavaScript technologies.

[PORTKEY-SRS-90] The backend shall be implemented using JavaScript/Node.js framework.

[PORTKEY-SRS-91] The database shall be MySQL or MongoDB based on scalability requirements.

[PORTKEY-SRS-92] The system shall follow RESTful API design principles for client-server communication.

[PORTKEY-SRS-93] All communications shall use HTTPS protocol for security and data protection.

[PORTKEY-SRS-94] The chatbot shall be implemented using Python-based ML libraries and frameworks.

2.7 Standards compliance

[PORTKEY-SRS-95] The system shall comply with IEEE 830-1998 standard for software requirements specification.

[PORTKEY-SRS-96] The application shall follow W3C web standards for HTML5, CSS3, and JavaScript.

[PORTKEY-SRS-97] The system shall comply with PCI DSS requirements for payment processing security.

[PORTKEY-SRS-98] The application shall adhere to GDPR and local data privacy regulations.

2.8 Software system attributes

2.8.1 Reliability

[PORTKEY-SRS-99] The system shall provide 99.5% availability during operating hours with automatic error recovery mechanisms.

[PORTKEY-SRS-100] The application shall implement comprehensive data backup and recovery procedures.

[PORTKEY-SRS-101] The system shall handle failures gracefully without data loss or corruption.

2.8.2 Security

[PORTKEY-SRS-102] The application shall implement robust user authentication and authorization mechanisms.

[PORTKEY-SRS-103] The system shall encrypt all sensitive data transmission using HTTPS protocol.

[PORTKEY-SRS-104] The application shall implement input validation to prevent SQL injection and XSS attacks.

[PORTKEY-SRS-105] The system shall maintain audit logs for all critical operations and transactions.

2.8.3 Maintainability

[PORTKEY-SRS-106] The system shall implement modular code architecture for easy maintenance and updates.

[PORTKEY-SRS-107] The application shall provide comprehensive API documentation for developers.

[PORTKEY-SRS-108] The system shall support version control using Git/GitHub for code management.

[PORTKEY-SRS-109] The application shall implement automated testing procedures for quality assurance.

2.8.4 Portability

[PORTKEY-SRS-110] The application shall provide cross-browser compatibility across modern web browsers.

[PORTKEY-SRS-111] The system shall implement responsive design for various screen sizes and devices.

[PORTKEY-SRS-112] The application shall support cloud deployment on multiple platforms (AWS, Azure, GCP).

2.8.5 Scalability

[PORTKEY-SRS-113] The system shall support horizontal scaling to handle increased user load.

[PORTKEY-SRS-114] The application shall implement caching mechanisms for improved performance.

[PORTKEY-SRS-115] The system shall support load balancing for distributed system architecture.

3. Verification

Verification tests shall be conducted to ensure all specified requirements are met. Testing shall include:

- **Unit Testing:** Individual component functionality verification
- **Integration Testing:** System component interaction validation
- **User Acceptance Testing:** End-user requirement satisfaction
- **Performance Testing:** Load and stress testing under various conditions
- **Security Testing:** Vulnerability assessment and penetration testing
- **Compatibility Testing:** Cross-browser and device functionality verification
- **Chatbot Testing:** ML model accuracy and response quality evaluation

Test cases shall be developed for each functional requirement with defined acceptance criteria and expected outcomes.

4. Supporting information

Additional supporting documentation includes:

- User Interface Design Mockups
- System Architecture Diagrams
- Database Schema Design
- API Documentation
- Deployment Guidelines
- User Manual and Help Documentation

5. References

- **IEEE Std 830-1998:** IEEE Recommended Practice for Software Requirements Specifications
- **ISO/IEC/IEEE 29148:2018:** Systems and software engineering requirements engineering
- **W3C Web Standards:** HTML5, CSS3, JavaScript specifications
- **PCI DSS:** Payment Card Industry Data Security Standard

- **GDPR:** General Data Protection Regulation guidelines
- **WCAG 2.1:** Web Content Accessibility Guidelines